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#include <stdio.h>
#include <stdlib.h>

int main()
{
    int request[50], n, head, disk;
    int i, j, temp, movement = 0;

    printf("Enter number of requests: ");
    scanf("%d", &n);

    printf("Enter request queue:\n");
    for(i = 0; i < n; i++)
        scanf("%d", &request[i]);

    printf("Enter initial head position: ");
    scanf("%d", &head);

    printf("Enter disk size: ");
    scanf("%d", &disk);

    /* Sort requests */
    for(i = 0; i < n - 1; i++)
    {
        for(j = i + 1; j < n; j++)
        {
            if(request[i] > request[j])
            {
                temp = request[i];
                request[i] = request[j];
                request[j] = temp;
            }
        }
    }

    printf("\nSeek Sequence: %d", head);

    /* Move towards right */
    for(i = 0; i < n; i++)
    {
        if(request[i] >= head)
        {
            movement += abs(head - request[i]);
            head = request[i];
            printf(" -> %d", head);
        }
    }
}

```

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/* Move to end of disk */
movement += abs((disk - 1) - head);
head = disk - 1;
printf(" -> %d", head);

/* Move towards left */
for(i = n - 1; i >= 0; i--)
{
    if(request[i] < head)
    {
        movement += abs(head - request[i]);
        head = request[i];
        printf(" -> %d", head);
    }
}

printf("\nTotal Head Movement = %d cylinders\n", movement);

return 0;
}

```

OUTPUT:

Enter number of requests: 8

Enter request queue:

98 183 37 122 14 124 65 67

Enter initial head position: 53

Enter disk size: 200

Seek Sequence:

53 -> 65 -> 67 -> 98 -> 122 -> 124 -> 183 -> 199 -> 37 -> 14

Total Head Movement = 331 cylinders